Climate Trends and Crop Yield Prediction

An Analytical Approach To Measure the Crop Yield based on Several Weather conditions for the Last 26 Years.

Presented by

B.STAT 2ND YEAR STUDENTS

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Contents

Abstract

The ongoing impact of climate change has introduced unprecedented challenges to global food security by influencing agricultural productivity. This project aims to explore the intricate relationship between climatic variables and crop yield over a span of 26 years. Utilizing a dataset that includes annual temperature, rainfall, soil moisture, and extreme weather events, we develop predictive models to estimate crop yield in response to shifting environmental conditions. The goal is to provide actionable insights that can assist policymakers and farmers in adapting to climate change through data-driven agricultural planning and sustainable farming practices.

1 Introduction

Agriculture is a climate-sensitive sector, and its productivity is directly affected by variations in temperature, rainfall, and the frequency of extreme weather events. With the increasing threat of climate change, it becomes imperative to understand how these factors influence crop yields. In recent decades, observable shifts in climatic patterns have had both direct and indirect effects on soil conditions, water availability, and the overall growing environment for crops.

This project investigates how climatic variables, like average temperature, annual rainfall, extreme weather events, and soil moisture impacts on crop yield. The dataset spans 25 years and reflects critical indicators of environmental stress and agricultural output. Through statistical analysis and machine learning models, we aim to predict crop yields and assess the sensitivity of agricultural systems to climate-related variables. The outcomes of this study are intended to support climate-resilient agricultural strategies and improve food production forecasting in the face of global environmental change.

2 Objective

The objective of this project is to investigate and model the effects of climate change on agricultural productivity, particularly focusing on crop yield. Beside that, this project aims to create predictive models to implement welfare policies.

Specific Objectives:

1. To analyze temporal trends in key climatic variables such as average temperature, annual rainfall, soil moisture, and the frequency of extreme weather events.
2. To study the correlation between climate indicators and crop yield, identifying which variables have the most significant impact on agricultural output.
3. To develop predictive models using statistical techniques that estimate crop yield based on climatic and environmental parameters.
4. To generate actionable insights for farmers and policymakers to help plan for climate-resilient agricultural practices.
5. To explore future projections of crop yield under changing climate conditions and suggest possible mitigation or adaptation strategies.
6. To contribute to the broader discourse on climate change and food security through data-driven research.

3 Dataset Description

The dataset used in this project is a simulated time-series dataset spanning from the year 2000 to 2024. It captures a set of climate-related variables that are hypothesized to influence agricultural productivity, particularly crop yield.
The dataset contains the following features:
Year (2000 - 2024):
Represents the time period over which the data was collected or simulated. This variable serves as the temporal index to observe long-term climate trends and changes in crop productivity.
Average Temperature (°C):
Indicates the average annual temperature in degrees Celsius for each year. Temperature is a critical climatic factor that affects crop growth phases such as germination, flowering, and maturation.
Annual Rainfall (mm):
Refers to the total amount of precipitation recorded annually, measured in millimetres. Adequate rainfall is essential for soil hydration and nutrient transport,

and its variation can significantly affect crop health and yield.
Extreme Weather Events (count):
Denotes the number of extreme weather occurrences (such as droughts, floods, storms, or heatwaves) in a given year. These events often lead to abrupt agricultural losses and stress on crop systems.
Soil Moisture (%):
Measures the percentage of moisture content in the soil, which directly influences plant water availability and root health. Soil moisture acts as a key intermediary between climatic conditions and crop performance.
Crop Yield (tons per hectare):
Represents the total output of crops harvested per unit area, measured in tons per hectare. This is the target variable of interest, and the goal of this study is to model and predict this variable based on the climatic inputs.

```r
data = read.csv("dataset.csv",header = TRUE)
dim(data)

## [1] 25  6

colnames(data)

## [1] "Year"                         "Average_Temperature_C"
## [3] "Annual_Rainfall_mm"           "Extreme_Weather_Events"
## [5] "Soil_Moisture_Percentage"     "Crop_Yield_ton_per_hectare"

colSums(is.na(data))

##                       Year      Average_Temperature_C
##                          0                          0
##         Annual_Rainfall_mm     Extreme_Weather_Events
##                          0                          0
##   Soil_Moisture_Percentage Crop_Yield_ton_per_hectare
##                          0                          0

sum(duplicated(data))

## [1] 0
```

4 Data Exploration

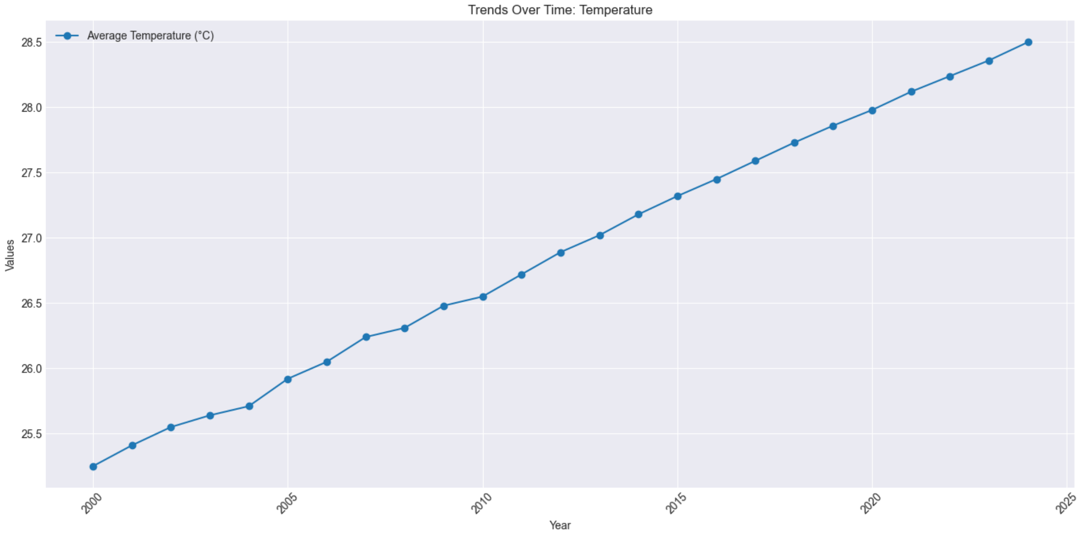
Data exploration is a crucial step in understanding the structure, patterns, and anomalies in the dataset before applying statistical analysis or predictive modeling. In this project, we perform both numerical and graphical exploration of the climate and crop yield data collected over the 25-year period from 2000 to 2024. Now we need to use some package for our statistical analysis. So let's Load this packages in this file.

```
library(readr)
library(corrplot)

## corrplot 0.92 loaded

library(corrplot)
library(ggcorrplot)

## Loading required package:  ggplot2

library(ggplot2)
library(caTools)
library(caTools)
library(tidyr)
```

At first, we look into the trends of the data with line plots.

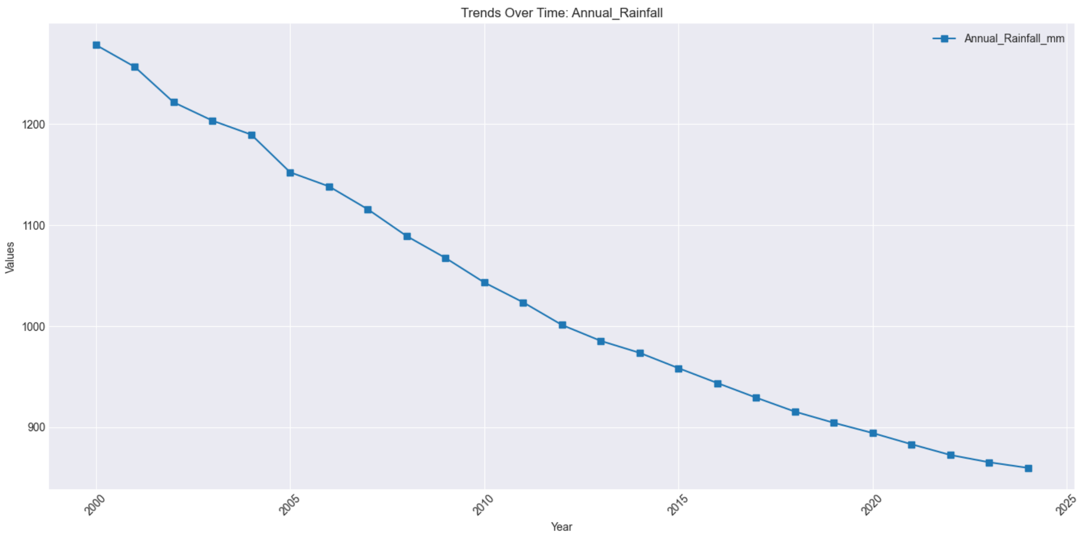
4.1 Average Temperature

A line graph was plotted to observe changes in temperature over time. This graph shows the effect of global warming in recent years.



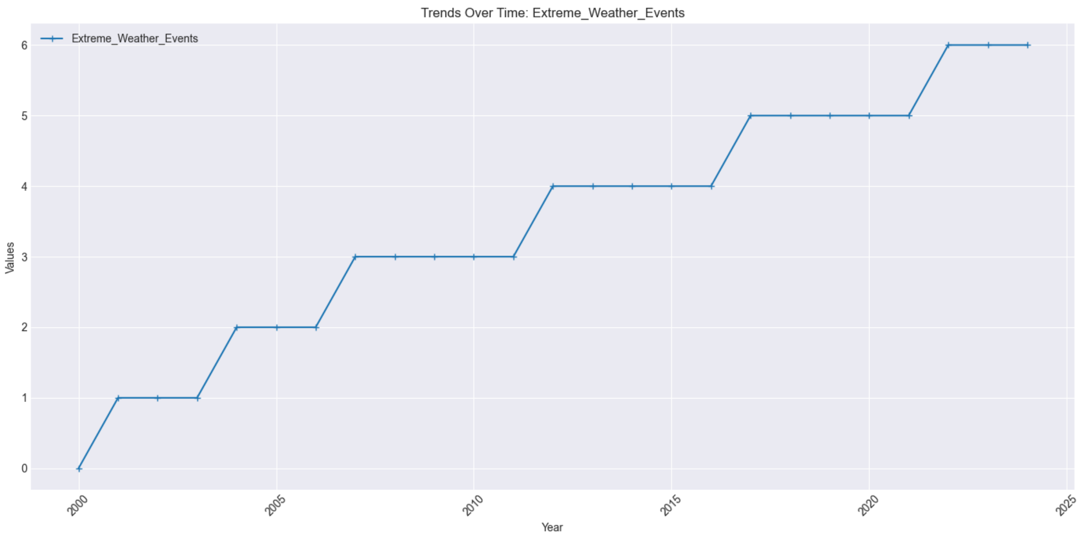
4.2 Annual Rainfall

Similar plots were generated for rainfall to detect patterns such as steep decrease of rainfall over time.



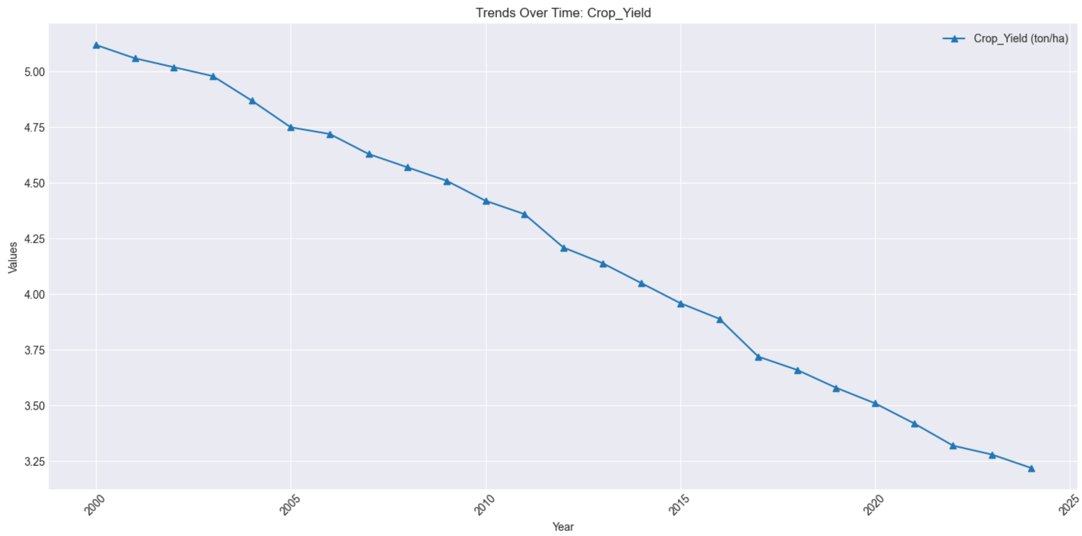
4.3 Extreme Weather Events

A bar chart showed the rising frequency of extreme weather events, which could indicate rising climate volatility.



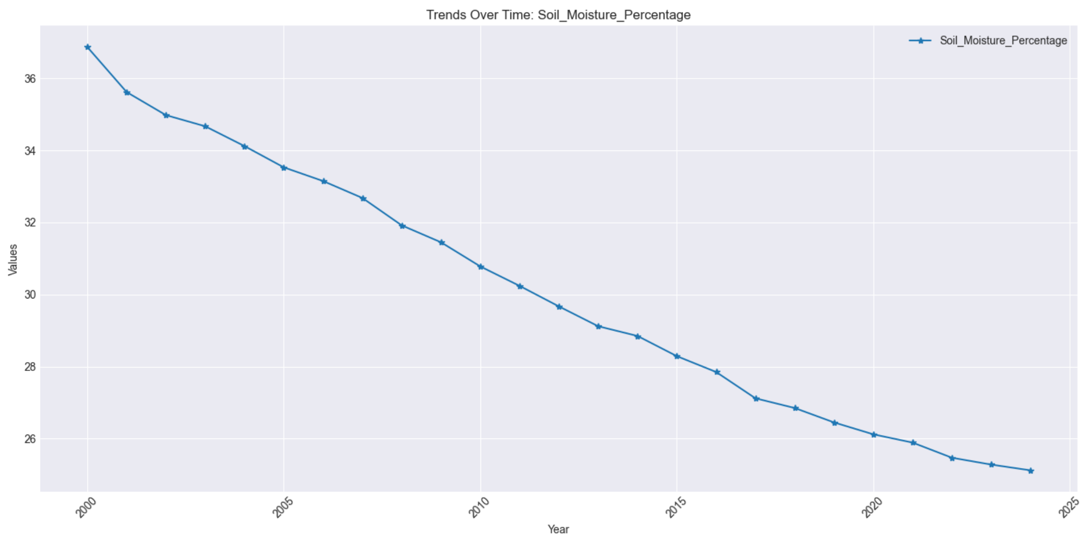
4.4 Crop Yield

Crop yield was plotted against time to visually assess if there's a downward trend, and if it aligns with any climate indicators.



4.5 Soil Moisture

Soil moisture trends help contextualize water availability for crops during various years.



5 Statistical Analysis

The statistical analysis phase investigates the relationships between climate variables and crop yield to identify key influencing factors. This includes computing correlations, conducting hypothesis testing, and visualizing inter-variable relationships.

5.1 Correlation Analysis

The Pearson correlation coefficient was calculated to evaluate the strength and direction of the linear relationship between crop yield and each climate variable.

Variables Analyzed:

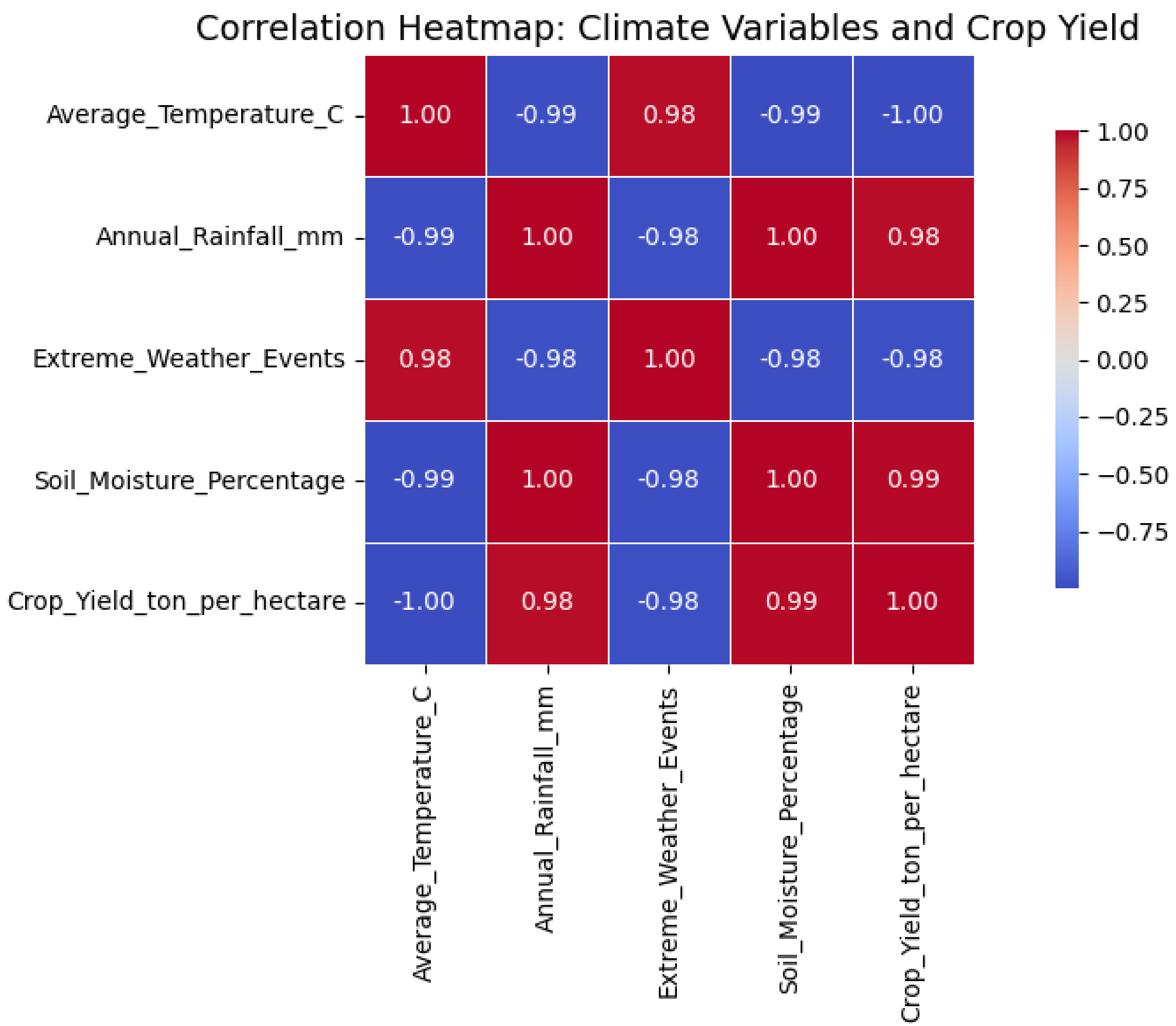
Crop Yield vs. Average Temperature
Crop Yield vs. Annual Rainfall
Crop Yield vs. Soil Moisture
Crop Yield vs. Extreme Weather Events

Objective:

To determine which climatic factors have the strongest positive or negative association with crop yield.

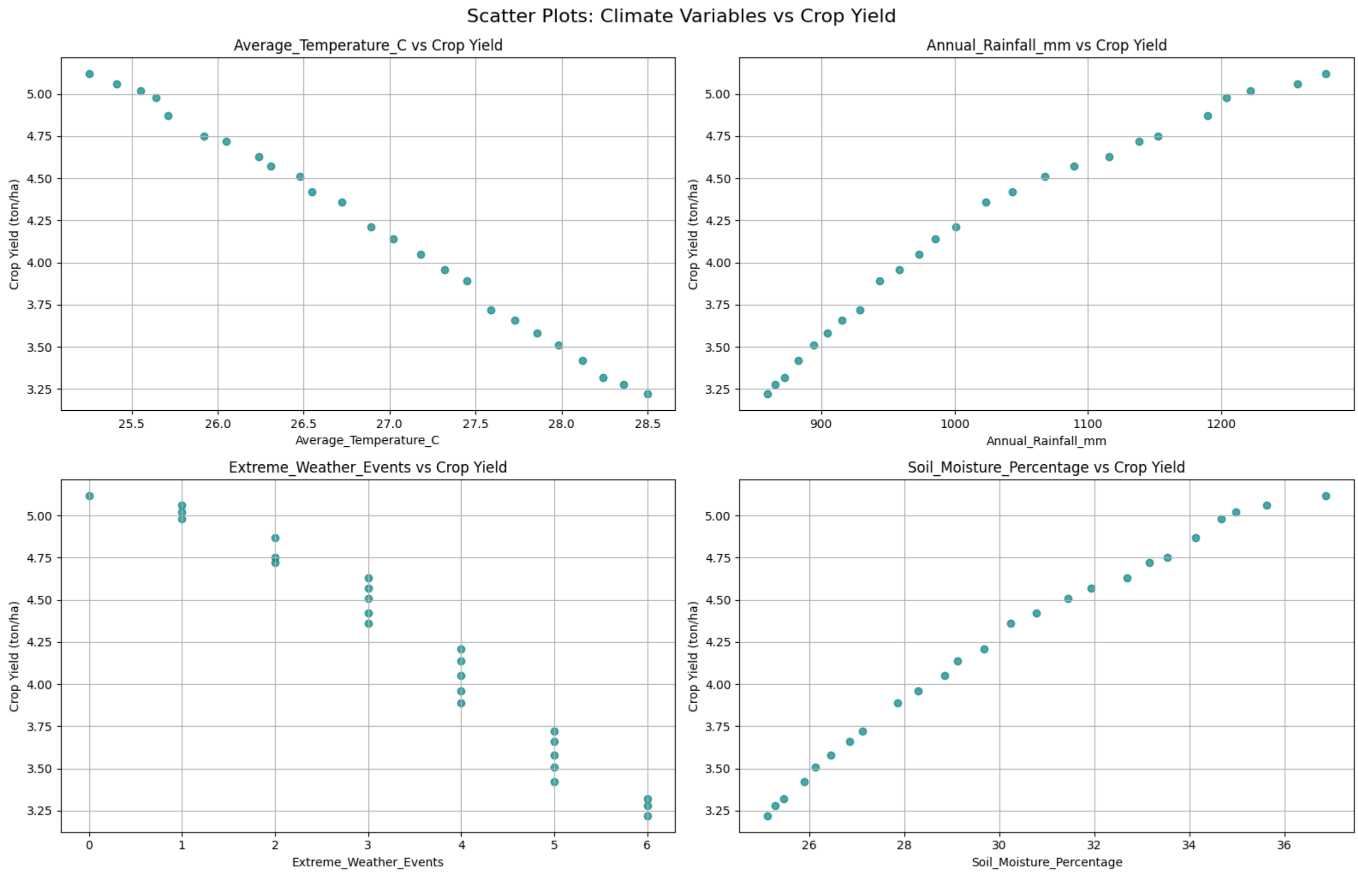
```
cor(data$Crop_Yield_ton_per_hectare, data$Average_Temperature_C)

## [1] -0.9982089

cor(data$Crop_Yield_ton_per_hectare, data$Annual_Rainfall_mm)

## [1] 0.9802736

cor(data$Crop_Yield_ton_per_hectare, data$Soil_Moisture_Percentage)

## [1] 0.9904854

cor(data$Crop_Yield_ton_per_hectare, data$Extreme_Weather_Events)

## [1] -0.9758542
```

Now let's visualise the correlation matrix with the heatmap.



5.2 Scatter Plot:

Scatter plots were used to visualize relationships between crop yield and each climate variable, allowing for visual confirmation of trends suggested by the correlation coefficients.



5.3 Hypothesis Testing

We want to fit a regression model of the weather condition variables and the crop yield. Before modelling, we need to test if the effect of the variables are statistically significant or not:

5.3.1 Test Used:

Analysis of variance test between the full model (involving all variables) and the reduced model (without involving the concerned variable). We test the null hypothesis

$$H_0 : \text{The variable has no significant effect}$$

against the alternative

$$H_1 : \text{The variable has significant effect}$$

```
full_model=lm(data$Crop_Yield_ton_per_hectare~data$Average_Temperature_C+dat
reduced_model=lm(data$Crop_Yield_ton_per_hectare~data$Annual_Rainfall_mm+dat
print(anova(full_model,reduced_model)['Pr(>F)'][2,1])
```

```
## [1] 8.261315e-09
```

5.3.2 Results:

1. **Effect of temperature:** The test finds that the effect of temperature is quite significant, with p value less than 10^{-7}.
2. **Effect of rainfall:** The test finds that the effect of rainfall is quite significant, with p value around 0.3%.
3. **Effect of natural calamities:** The effect of unnatural weather events is not that significant, with p value around 66.22%
4. **Effect of soil moisture:** The effect of soil moisture is moderately significant, with p value 7.52%

5.4 Conclusion:

The analysis confirmed that certain climate variablesâĂŤparticularly rainfall and soil moisture play a significant role in determining crop yields. These findings offer a strong foundation for building predictive models and making data-driven agricultural recommendations.

6 Predictive Modelling:

This section focuses on developing statistical models to predict crop yield based on climate variables. By applying regression techniques, we aim to quantify the influence of variables such as temperature, rainfall, soil moisture, and extreme weather events on agricultural productivity.

6.1 Objective:

To build a predictive model that accurately estimates crop yield (tons per hectare) using climatic variables as predictors:
Average Temperature (°C),Annual Rainfall (mm),Extreme Weather Events,Soil Moisture (%)

6.2 Methodology:

6.2.1 Linear Regression Analysis

A Multiple Linear Regression (MLR) model was employed to predict crop yield as a linear combination of multiple climate factors. This technique helps in understanding the direction and magnitude of each variable's effect on the yield. Using the standard multiple linear regression model, we get a satisfactory fit.

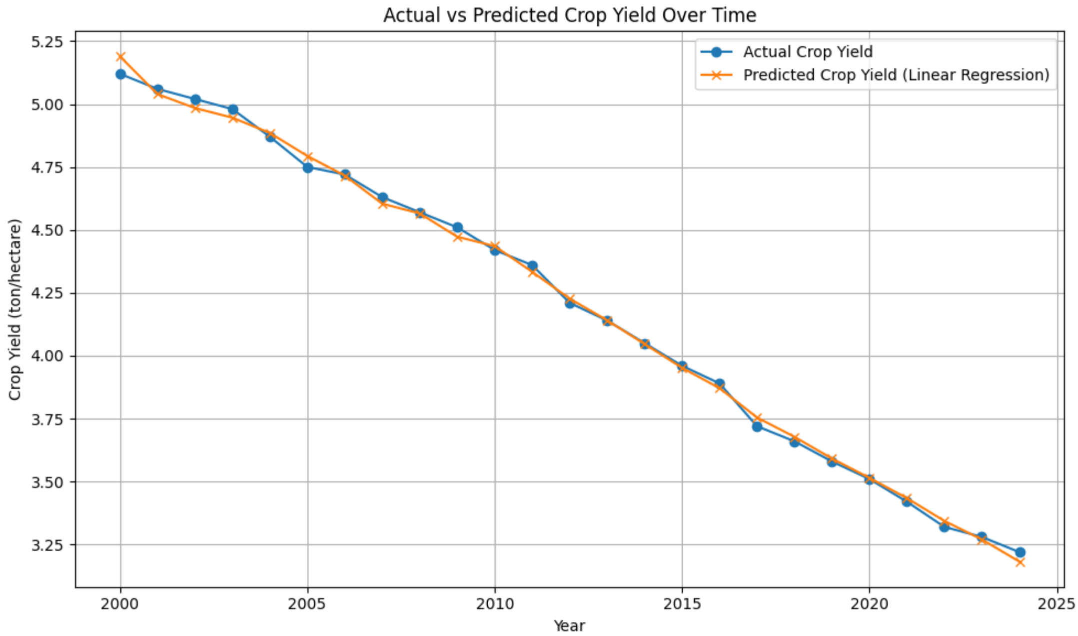
```
summary(full_model)

##
## Call:
## lm(formula = data$Crop_Yield_ton_per_hectare ~ data$Average_Temperature_C
##     data$Annual_Rainfall_mm + data$Soil_Moisture_Percentage +
##     data$Extreme_Weather_Events)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.069314 -0.015521  0.003547  0.020495  0.037698
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    21.5228509  2.3860729   9.020 1.74e-08 ***
## data$Average_Temperature_C     -0.6361841  0.0673915  -9.440 8.26e-09 ***
## data$Annual_Rainfall_mm        -0.0022519  0.0006677  -3.373  0.00303 **
## data$Soil_Moisture_Percentage   0.0707658  0.0377019   1.877  0.07519 .
## data$Extreme_Weather_Events    -0.0084750  0.0191129  -0.443  0.66222
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02962 on 20 degrees of freedom
## Multiple R-squared:  0.9981, Adjusted R-squared:  0.9977
## F-statistic:  2598 on 4 and 20 DF,  p-value: < 2.2e-16
```

Hence, if we call crop yield Y, and the effects temperature, rainfall, unprecedented events and soil moisture X_1, X_2, X_3, X_4 respectively, then the fitted model is

$$Y = (-0.636184)X_1 - (0.002252)X_2 - 0.008475X_3 + 0.070766X_4 + 21.522851$$

6.2.2 Comparison

Here is the visualisation of fitted and actual data.



6.3 Conclusion of Predictive Modelling

The regression model successfully identifies the key drivers of crop productivity. It provides an interpretable and quantifiable approach to forecast future yields based on climate projections. Such models are valuable for farmers, agricultural planners, and policymakers to make informed, data-driven decisions for climate-resilient agriculture.

7 General Recommendations

Based on the statistical findings and predictive analysis, several practical and policy-oriented recommendations can be made to enhance agricultural productivity and resilience in the face of changing climate conditions.

7.1 Recommendations for Farmers

Adopt Climate-Smart Agricultural Practices

Sustainable farming techniques such as crop rotation, conservation tillage, and organic composting should be implemented to ensure soil health and moisture retention. Drought-resistant or flood-tolerant crop varieties should be used to endure adverse weather conditions.

Efficient Water Management

Drip or sprinkler irrigation systems should be invested in to enhance water use efficiency, particularly in areas experiencing erratic rainfall. Rainwater should be harvested and stored during periods of high rainfall for use during dry spells.

Monitor and Respond to Weather Events

Real-time weather forecasting tools should be utilized to guide sowing, irrigation, and harvesting decisions. Crop types should be diversified or planting should be staggered to minimize the risk of complete yield loss during extreme weather events.

7.2 Recommendations for Policymakers and Agricultural Planners

Strengthen Early Warning Systems

Access to early warnings for droughts, floods, and storms should be improved to enable farmers to prepare and reduce losses. Climate forecasts should be disseminated in local languages through accessible platforms (e.g., SMS, radio, or mobile apps).

Invest in Agricultural Infrastructure

Irrigation canals, water storage facilities, and weather monitoring stations should be developed in rural areas. Local institutions and cooperatives that provide training and resources on climate-adaptive farming should be supported.

Promote Data-Driven Decision Making

Statistical and predictive models should be integrated into national agricultural policy frameworks to guide subsidies, insurance programs, and crop planning. The use of data analytics in agricultural extension services should be encouraged to offer customized support to farmers.

7.3 Continuous Monitoring and Feedback

A system for regularly monitoring climate and crop data should be established, with models updated accordingly. Farmers should be engaged in participatory research to validate model predictions and adjust recommendations at the grass-roots level.

8 Conclusion

This project highlights how climate factors like rainfall, soil moisture, and temperature impact crop yields. Using predictive modeling, strong links were found between climate trends and agricultural output. To enhance climate-resilient farming, key measures include forecast-based planning, real-time climate data systems, adaptive crop models, and collaborative research. Policy integration and climate-risk insurance can further protect farmers. Data-driven strategies are essential for adapting to climate uncertainty and ensuring food security through informed agricultural decision-making and resource optimization.